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The Unactivated Biomedical Technology

Diffusion of Generic Drugs Beyond the Prescription

Docket	FDA-2023-N-0119
RFI	Input on FY 2027 GDUFA Science and Research Initiatives
Subject	Generic-drug diffusion / GDUFA science & research (FY 2027)
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The core argument

A biopharmaceutical technology has not reached the patient when it is approved, dispensed, or prescribed. It reaches the patient only when it is **activated through label-consistent use**. Between prescribing and that activation, a large share of approved — and already paid-for — therapeutic value is lost.

FDA’s question

FDA asks what additional research could enhance public access to high-quality, safe, and effective generic products. One gap is hidden in the meaning of access itself: a biopharmaceutical technology has not reached the patient when it is approved, dispensed, or prescribed. It reaches the patient only when it is activated through label-consistent use.

1. The object is misidentified.

A prescription drug is a high-cost biomedical technology. Unlike software, it cannot deliver utility by being acquired. It requires biological activation — it must enter the body under the dose, timing, duration, and regimen conditions specified in its labeling to alter disease physiology safely and effectively. Medicine left in the bottle cannot help. A prescription transfers the artifact; label-consistent use activates the technology.

Between prescribing and label-consistent use, a large share of the approved therapeutic value is lost. This loss is conventionally placed in the category of medication "non-adherence," subdivided into intentional and unintentional behavior. That framing names an outcome without identifying the system that produced it — and therefore misdirects the remedy.

2. The gap in network science.

Network science treats high-cost, risky technologies as complex contagions — adoption requires reinforcement from multiple independent sources rather than a single exposure (Centola & Macy, 2007), building on earlier threshold models of collective behavior (Granovetter, 1978).

But pharmaceutical diffusion research maps the supply side — physician prescribing networks — and treats the prescription as the adoption endpoint (Coleman, Katz & Menzel, 1966; Van den Bulte & Lilien, 2001). This measures distribution of the product, not deployment of the technology.

A biopharmaceutical technology is undiffused until it lands in label-consistent use. The consumer side — where biological activation actually occurs — remains largely unstudied.

3. What the model predicts.

Complex contagion requires reinforcement from nodes with sameness: people using the same technology for the same indication. On this model, clinicians and pharmacists — who are not themselves using the drug — transmit information (simple contagion) but cannot supply the clustered, redundant reinforcement complex contagion requires (Centola, 2010: clustered networks outperform sparse ones for behaviors needing social reinforcement). The same structural limit applies to notifications and general digital companions.

This yields a falsifiable prediction consistent with observed data: label-consistent use fails most in stigmatized, socially invisible conditions — precisely where patients have no observable nodes with sameness in their natural networks. Under an individual-behavior model, that pattern has no explanation.

4. Consistency is already measurable — without surveillance.

FDA has established that consistency of use can be assessed through indirect, biomarker-anchored endpoints. Its 2018 prescription drug-use-related software (PDURS) framework describes exactly this: evidence that a dose-tracking or reminder app improves patient compliance — shown through a reduction in the validated biomarker hemoglobin A1c (HbA1c) compared to the drug alone — would be sufficient to support a labeling claim (Docket FDA-2018-N-3017; 83 FR 58577, Nov. 20, 2018). Consistency is therefore measurable without self-report and without monitoring individuals. The measurement precedent exists; the mechanistic model does not.

5. Why OGD, and why now.

Generic products cannot differentiate on the molecule without losing therapeutic equivalence. This structurally suppresses investment in realized use — leaving the largest remaining source of public-health value in generics unaddressed. That value is already approved and already paid for; it simply does not arrive. FDA need not wait for network science to correct the boundary of pharmaceutical diffusion.

6. Recommendation — proposed FY 2027 research questions.

Building on the FY 2025 priority Expand the Use of AI/ML Tools, including "improving the use of real-world evidence for post-market surveillance … and for evaluating the impact of generic drugs on public health," we recommend research addressing:

Topology. What patient-network structures predict initiation, persistence, and label-consistent use after dispensing?

Reinforcing node. What constitutes a relevant reinforcing node for a given product, indication, and regimen?

Clustering without surveillance. Can privacy-preserving methods identify and cluster nodes with clinically relevant sameness without exposing the diagnosis or treatment status of any individual?

Label integration. How can a networked activation system be integrated with FDA-approved labeling, consistent with PDURS, where consistency of labeled use is essential to safe and effective use — without altering the indication?

Endpoints. What biomarker-anchored endpoints validate label-consistent use across generic classes, without self-report or monitoring?

Attribution. Can postmarket data distinguish failure of product quality from failure of biological activation, and quantify the resulting loss of public-health value?

7. Conclusion.

In closing remarks at the 2025 Generic Drugs Forum, Kendra Stewart, R.Ph., Pharm.D. — Deputy Director of Operations, Office of Generic Drugs — named the goal: to bring therapies to patients more efficiently. A therapy is not brought to a patient when the prescription is written. It is brought to the patient when the technology is activated through label-consistent use — and its approved therapeutic value becomes biologically available.

We would welcome the opportunity to contribute to this research agenda.

References

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